

MATERIAL SAFETY DATA SHEET (MSDS)

SECTION 1: PRODUCT AND COMPANY IDENTIFICATION

Product Name: Bitumen 200/300

Chemical Name: Asphalt, Petroleum Asphalt

CAS Number: 8052-42-4

Product Use: Road construction, waterproofing, roofing, industrial applications

Company Name: ATDM Co.

SECTION 2: HAZARDS IDENTIFICATION

GHS Classification:

- Skin Irritation (Category 2)
- Eye Irritation (Category 2B)
- Carcinogenicity (Category 2)
- Specific Target Organ Toxicity – Repeated Exposure (Category 2)

Signal Word: WARNING

Hazard Statements:

- May cause skin and eye irritation.
- May cause respiratory irritation if inhaled in fumes.
- Suspected of causing cancer through prolonged exposure.

Precautionary Statements:

- Avoid breathing fumes or vapors.
 - Wear protective gloves and safety glasses.
 - Use only in well-ventilated areas.
 - Wash thoroughly after handling.
 - Avoid prolonged exposure to high temperatures.
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SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Component CAS Number Concentration

Bitumen 8052-42-4 100%

SECTION 4: FIRST AID MEASURES

Inhalation: Move to fresh air. Seek medical attention if breathing difficulties persist.

Skin Contact: Wash affected area with soap and water. If irritation develops, seek medical advice.

Eye Contact: Flush eyes with plenty of water for at least 15 minutes. Remove contact lenses if present. Seek medical attention if irritation persists.

Ingestion: Not a likely route of exposure. If swallowed, seek medical attention immediately.

SECTION 5: FIRE-FIGHTING MEASURES

Suitable Extinguishing Media: Dry chemical, carbon dioxide (CO₂), foam, or water spray.

Unsuitable Extinguishing Media: Do not use direct water jet as it may spread the fire.

Fire Hazards: Bitumen is not highly flammable but can ignite at high temperatures, releasing toxic fumes.

Firefighting Instructions: Firefighters should wear self-contained breathing apparatus (SCBA) and protective clothing.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions: Use appropriate protective equipment. Avoid inhalation of vapors and contact with skin.

Environmental Precautions: Prevent spillage from entering drains or waterways.

Spill Cleanup Methods: Allow bitumen to cool and solidify. Use mechanical means to remove. Dispose of in accordance with local regulations.

SECTION 7: HANDLING AND STORAGE**Handling:**

- Avoid direct contact with skin and eyes.
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- Use only in well-ventilated areas.
- Do not overheat to avoid fume production.

Storage:

- Store in a dry, well-ventilated area away from heat and open flames.
- Keep containers tightly closed when not in use.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**Exposure Limits:**

- OSHA PEL: 5 mg/m³ (fumes)
- ACGIH TLV: 0.5 mg/m³ (fumes)

Personal Protective Equipment (PPE):

- **Respiratory Protection:** Use NIOSH-approved respirator if fumes are present.
- **Hand Protection:** Wear heat-resistant gloves.
- **Eye Protection:** Use safety goggles or face shield.
- **Skin Protection:** Wear protective clothing.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Property	Value
Appearance	Black, viscous solid
Odor	Petroleum-like
Boiling Point	> 470°C
Flash Point	> 230°C
Specific Gravity	1.01 – 1.05
Solubility	Insoluble in water

SECTION 10: STABILITY AND REACTIVITY

Stability: Stable under normal conditions.

Hazardous Reactions: None under normal processing conditions.

Incompatible Materials: Strong oxidizing agents.

Hazardous Decomposition Products: Carbon monoxide (CO), carbon dioxide (CO₂), and other hydrocarbons when heated.

SECTION 11: TOXICOLOGICAL INFORMATION

Routes of Exposure: Inhalation, skin contact, eye contact.

Acute Toxicity: Low acute toxicity; prolonged exposure to fumes may cause irritation.

Chronic Effects: Prolonged or repeated exposure to bitumen fumes may cause skin disorders and respiratory issues.

Carcinogenicity: Bitumen is classified as **possibly carcinogenic to humans (IARC Group 2B)** due to polycyclic aromatic hydrocarbons (PAHs) in fumes.

SECTION 12: ECOLOGICAL INFORMATION

Environmental Impact:

- Bitumen is not readily biodegradable.
- Spills may cause environmental contamination.
- Avoid release into waterways.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal: Dispose of in accordance with local, regional, and national regulations.

Container Disposal: Empty containers may contain residues. Handle with care.

SECTION 14: TRANSPORT INFORMATION

UN Number: Not classified as hazardous for transport.

Proper Shipping Name: Petroleum Asphalt

Transport Hazard Class: Not regulated.

Packing Group: Not applicable.

SECTION 15: REGULATORY INFORMATION

Global Regulations:

- **OSHA:** Hazardous under OSHA Hazard Communication Standard (29 CFR 1910.1200).
 - **IARC:** Group 2B – Possible Carcinogen.
 - **EU REACH:** Listed.
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SECTION 16: OTHER INFORMATION

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Prepared By: ATDM Co.